Screen agent for API-less systems: design

Frontier Bogotá 2026 · Challenge 1 · Team design

1. Problem

Turn a natural-language request into **exactly** the right outcome inside a legacy business app the agent has never seen, using only pixels and mouse/keyboard over VNC, in under 5 minutes. "Right" means the expected record changes and nothing else: no extra edits, no duplicates, an honest final state (`fallida` when the task is ambiguous, impossible or dangerous) and a summary the grader can parse. The hard part is not clicking. It is **generalizing to unknown UIs and knowing when to stop**.

2. Assumptions

1. Evaluation apps are reached like the practice app (noVNC, 1280×800, same endpoints) and are typical CRUD tools: menus, lists, search/filters, forms, modals, confirmations, exports.
2. The screen is the only source of truth. No DOM, no API, no network from the app.
3. "Today" is the date shown by the app, not the server clock.
4. The hidden set size is unknown, so the cost per task must stay far below the US$5 reserve (target ≤ US$0.10).
5. The UI can change during the day, so nothing about a specific app is hardcoded.

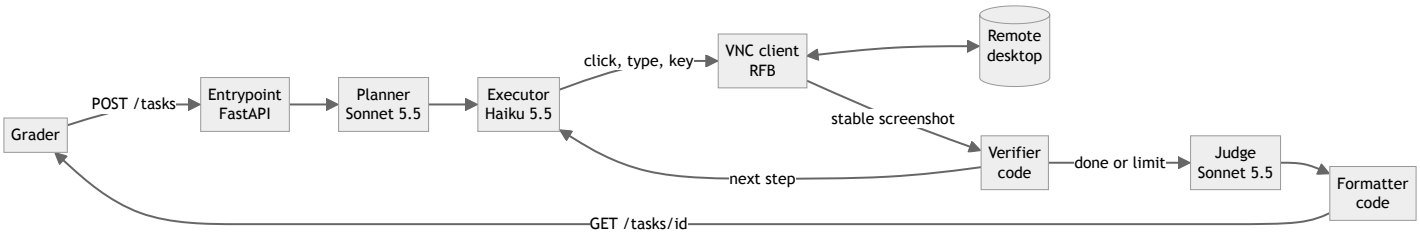
3. Hard cases and how we handle them

Case	Handling
Ambiguous entity (two "Ana Torres")	Never modify. Report `fallida` naming the distinguishing data (IDs, documents)
Impossible or dangerous request	`fallida` with the reason. Destructive actions outside the request are refused
Questions (counts, totals)	Use the app's own filters and totals, then `respuesta: <number first>` ; two values joined with `;`
Duplicates	Search before creating; never resubmit a form without checking the list first
Slow repaint / ignored fast clicks / text concatenation	Code waits for a stable frame, holds each click 150 ms, selects all before typing
Running out of time	Hard stop at 4:30, report, and stop touching the desktop

4. Out of scope (and why)

Persistent per-app memory across tasks (stale after a UI change, and a wrong memory is worse than none) · multi-app workflows · CAPTCHAs · fine-tuning a model (no time to collect data that generalizes).

5. Architecture



Code owns everything that must be exact; models own what needs judgment. **Planner** reads the task and first screen, flags ambiguity and danger. **Executor** proposes one action per screen. **Verifier** checks that the action changed the screen and enforces limits. **Judge** sees the final screen and action log and decides `terminada` / `fallida` . **Formatter** guarantees the grader format and removes secrets. Every step is logged with a timestamp and a screenshot.

6. Decisions

Decision	Rejected alternative	Why
Vision model reads raw screenshots	OCR + text-only model	OCR loses layout (which label owns which input, icons, checkboxes). OCR stays as an optional hint if misreads dominate failures
Haiku 5.5 executes each step (est. 25 steps: ~US$0.06/task incl. judge, ~83 tasks per US$5)	Sonnet 5.5 as executor (~US$0.31/task, ~16 tasks per US$5)	Sonnet is about 20× the price and slower per step, so fewer steps fit in 5 min. A measured bake-off decides
Sonnet only where judgment matters (plan, final verdict, escalation)	One cheap model for everything	False success and wrong refusals are the costliest errors; two strong calls per task are cheap insurance
Model proposes, code verifies	Trust the model's own report	Grader penalizes false success; screen-diff and limits catch silent failures
Direct RFB client over websocket	Headless browser running noVNC	Fewer layers and exact click timing. noVNC stays as fallback
Fresh exploration per task	Persistent memory per app	Robust to UI changes; correctness beats speed
Railway, always on (Docker)	Laptop + tunnel	The contract requires uptime through demo day

7. How we measure (before building)

- **One command** `uv run python -m evals.run` resets the desktop, sends each case to our entrypoint, calls `/__check` , verifies `estado` and summary format, and prints success rate, cost per task, latency and failure type.
- **Cases:** the 5 given + ~15 of ours (rephrasings and new tasks per type: create, edit, question, export, ambiguous, impossible, destructive, relative dates, two-value answers).
- **Generalization:** a **holdout app** with another domain and layout that the agent never sees during tuning, plus a check that fails if practice-app names appear in agent code or prompts.
- **Targets:** ≥ 80% own set, ≥ 60% holdout, 0 destructive actions, ≤ US$0.10 and p95 < 4 min per task. The 3 hidden-set runs are milestones: skeleton, after the incident, final.

8. Risks and monitoring

Risk	Detection	Response
UI changes mid-day (15:30)	"No effect" and "stuck" alarms; eval baseline at 15:00 and rerun	Per-case diff shows what broke; fix and document in README
Misclicks by the small model	Repeated no-effect actions	Escalate that step to Sonnet after 2 failures
Budget overrun	Cost tracked per call	Per-task cost cap; stop and report